

---

## MP4 + 5: Fragment Shader: Phong Illumination

**Due Time: Refer to the course web-site**

---

### Objectives

Verify that we understand and are able to work with:

- Unity specific skills:
  - Capable of working with and programming scripts to support per-update shader parameters
  - Using multiple arrays to implement an array of struct-construct
  - Passing of array of information to shaders
  - Differentiate between Main Camera and Editor Camera in illumination computation
  - Interactive controls of shader parameters
- Shader and relevant general concepts:
  - Details of Phong Illumination Model
  - Light attenuation: both distance and angular
  - Realtime computation of the Phong Illumination Model

### Approach

This is a straightforward assignment: we will implement the normalized BlinnPhong illumination model:

$$I = k_a + \sum_{i=0}^3 A_i D_i I_i \left( k_d (\hat{N} \cdot \hat{L}_i) + k_s \left( \frac{s+2}{2\pi} \right) (\hat{N} \cdot \hat{H}_i)^s \right)$$

Where:

- The camera:
  - $\hat{V}$ : the view vector, from the camera to the current point being illuminated
- Per object material
  - $k_a$   $k_d$   $k_s$ : are ambient, diffuse, and specular intensity and colors
  - $s$ : Phong specularity exponent
- Geometry:
  - $\hat{N}$ : the normal at the position being illuminated
  - $\hat{L}_i$ : the vector from position to light source-i
  - $\hat{H}_i$ : the halfway vector for light source-i, recall that this vector is half-way between  $\hat{V}$  and  $\hat{L}_i$
- Light source: there are at least four
  - $I_i$ : light intensity and color of light source-i
  - $A_i$ : angular attenuation defined by two cone angles
    - If inside the inner cone angle,  $\theta$ : 1.0
    - If outside of the outer cone angle,  $\varphi$ : 0.0
    - If in between, angle of  $\alpha$  with DropOff being a user define parameter:  
 $\text{smoothstep}(0.0, 1.0, \text{pow}(\frac{\cos \alpha - \cos \varphi}{\cos \theta - \cos \varphi}, \text{DropOff}))$
  - $D_i$ : distance attenuation defined by two distances: n and f
    - If closer than n: 1.0
    - If further than f: 0.0
    - If in between:  
 $\text{smoothstep}(0.0, 1.0, 1.0 - (n*n)/(d*d));$

The implementation of the above equation will be accomplished in two MPs:

- MP4:
  - Build general infrastructure, with multiple light sources
  - Support Ambient, Diffuse, and Texture
  - Support point and directional light types

$$I = k_a + \sum_{i=0}^3 I_i (k_d (\hat{N} \cdot \hat{L}_i))$$

- MP5: Completes the equation with:
  - Support specular component
  - Support Spot Light
  - Support light attenuation: distance and angular

## Hints

By now, we are relatively comfortable with working with shader code, and, should have a good sense of how variables in the shader can be updated from the editor either via the shader-properties in Material-UI or via customized scripts attached to GameObjects.

A sample [TemplateScene Unity project](#) is provided for your reference to help you begin MP4. This is based on my solutions. You are free to based your implementation on this project, or, ignore this and design your own solution. The following are some details of this template project.

- **Scene:**
  - Contains a teapot, two flattened spheres (as references for simple shapes), a prop\_barrel, ground plane, and a series of four spheres in the TestSpheres-group. These test spheres are meant for testing of specularity in MP5.
  - All geometries has the incomplete **552\_M2\_Material/Shader** and has the **MaterialLoader** component.
  - You need to complete the **552\_M2\_Shader**.
  - **MaterialLoader:** this component is meant to load appropriate per-object material information to the shader.
- **Light sources:**
  - Four spheres are defined as children of the SceneControl object. The Transforms of these objects are meant to be the light sources. **Make sure** that the SceneControl Transform is identity, otherwise, the Transforms of the lights will be wrong.
  - The **LightSource** component is attached to each of these spheres. This script defines parameters for simple visualization of the light types (when the **Debug** flag is switched on). You should modify this script to allow users to update other relevant parameters of a light source. It is safe to ignore both the Start() and Update() functions of this script (defined for visualization of the light).
  - **LightsLoader:** Is meant to load light source information to the Shader. During each update, the LightSourceSetLoader() function will be called (refer to SceneControl discussion), and this function should load the light source states to the shader.
- **Material:** As mentioned, the MaterialLoader script can load the material information to the shader.
- **Scene Control:** The **SceneControl** script is attached to the SceneControl game object. This script:
  - Defines the rendering flag: global switches that controls what to show during rendering
  - Has the array of four light sources referencing the four light source objects
  - During each update, this script should load the rendering flag and the light source state (by calling the LightsLoader.LightSourceSetLoader()) to the shader.

## Credit Distribution

You will loose credits if MP4 functionality is absent from MP5. E.g., In your MP5, if your light cannot be configured as Point or Directional light type, you will loose 15 points.

|                                                     | MP4 | MP4 Credits<br>Received | MP5 | MP5 Credits<br>Received |
|-----------------------------------------------------|-----|-------------------------|-----|-------------------------|
| <b>1 Unity Scene: based on the given template</b>   |     |                         |     |                         |
| At least 8 objects                                  | 2   | 2                       |     |                         |
| Group of 4 spheres with: N=0, 2, 20, 100            |     |                         | 6   | 6                       |
| All with own instance of material                   | 2   | 2                       |     |                         |
| Resting on a plane                                  | 2   | 2                       |     |                         |
| Game Window Camera Manipulation                     | 5   | 5                       |     |                         |
| <b>2 Light Sources</b>                              |     |                         |     |                         |
| At least 4 light sources                            | 12  | 12                      |     |                         |
| On/Off control                                      | 5   | 5                       |     |                         |
| Configured as: Point or Directional                 | 15  | 15                      |     |                         |
| Configured as: Point, Directional, or Spot          |     |                         | 18  | 18                      |
| Distance attenuation: Point + Spot                  |     |                         | 12  | 12                      |
| Angular attenuation                                 |     |                         | 15  | 15                      |
| All lights on: two point and two directional        | 8   | 8                       |     |                         |
| All lights on: at least 1 spot on a distinct object |     |                         | 6   | 6                       |
| Interactive parameter setting from editor           | 14  | 14                      |     |                         |
| <b>3 Materials</b>                                  |     |                         |     |                         |
| Ka + Ambient Color                                  | 4   | 4                       |     |                         |
| Kd + Diffuse Color                                  | 4   | 4                       |     |                         |
| Diffuse Color Texture                               | 1   | 1                       |     |                         |
| KS + Specular Color + Specularity                   |     |                         | 15  | 15                      |
| Interactive parameter setting from editor           | 14  | 14                      |     |                         |
| <b>4 Scene Control</b>                              |     |                         |     |                         |
| View from MainCamera or Editor Camera               |     |                         | 12  | 12                      |
| Texture On/Off                                      | 2   | 2                       |     |                         |
| Ambient On/Off                                      | 2   | 2                       |     |                         |
| Diffuse On/Off                                      | 2   | 2                       |     |                         |
| Specular On/Off                                     |     |                         | 2   | 2                       |
| Distance attenuation On/Off                         |     |                         | 4   | 4                       |
| Angular attenuation On/Off                          |     |                         | 4   | 4                       |
| <b>5 Submission</b>                                 |     |                         |     |                         |
| Proper unity version + project name                 | 3   | 3                       | 3   | 3                       |
| Four folders in submission                          | 3   | 3                       | 3   | 3                       |

|              |            |            |            |            |
|--------------|------------|------------|------------|------------|
| <b>Total</b> | <b>100</b> | <b>100</b> | <b>100</b> | <b>100</b> |
|--------------|------------|------------|------------|------------|

Each of MP4 and MP5 counts 10% towards your final grade for this class. Together, the two MPs counts 20%